Implicit and Explicit Phonetic Convergence to Voice Quality

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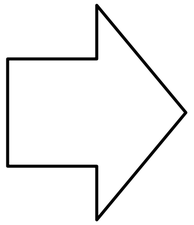
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RESEARCH QUESTIONS

1. Is voice quality a feature to which speakers will converge?
2. How do implicit instructions to repeat and explicit instructions to imitate a talker affect the degree of convergence?
3. How does convergence to voice quality interact with both speaker and listener gender?

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ANSWERS TO THESE QUESTIONS

1. Yes

2. Different acoustic parameters seem to behave differently in different reproduction conditions

PHONETIC CONVERGENCE

-People tend to speak like those around them
  -e.g. children acquiring their first language

-A similar-if-not-related phenomenon:
  -listeners adapt their own speech relative to the speech they perceive (Nguyen and Delvaux, 2015; Pardo, 2012, 2013)

PHONETIC CONVERGENCE

Phonetic convergence (or imitation, or accommodation):

adaptive processes result in speakers' adoption of features from the speech of interlocutors.

-in natural conversations
-in spontaneous interaction
-in collaborative lab tasks (Giles et al.,1973; Natale, 1975; Pardo, 2007)
-in less-realistically-social, non-interactive tasks, like word shadowing (Goldinger, 1998; Shockley et al., 2004)

PHONETIC CONVERGENCE

The resulting proposal is that convergence results from automatic processes reflecting a close link between perception and production
(Delvaux and Soquet, 2007; Babel, 2010; Dufour and Nguyen, 2013)

(But in that everyone doesn't produce phonetically identical speech, there are limits to this phenomenon, generally)

CONVERGENCE TO ACOUSTIC PROPERTIES

There are a number of acoustic parameters that we generally expect people to converge to in shadowing tasks:

-Vowel Duration (Hargreaves, 1960; Webb, 1970)
-F0 (Goldinger, 1997)
-F1, F2 (Tilsen, 2009; Babel, 2012)
-VOT (Shockley et al., 2004; Nielsen, 2007, 2011),

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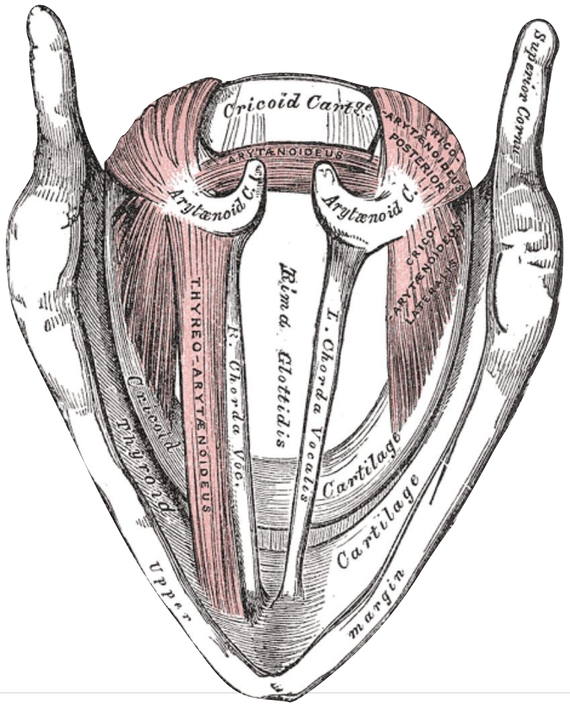
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The first goal of this study is to test whether acoustic correlates of voice quality behave similarly

VOICE QUALITY

4 phonation types considered in this study:

- Modal
- Breathy
- Falsetto
- Creaky



ACOUSTIC MEASUREMENTS

For the purposes of this study, I assess voice quality through acoustic correlates of the difference in amplitude between the first and second harmonic (H1-H2), Cepstral Peak Prominence (CPP), and f0

H1-H2 is correlated with glottal spread/constriction;
CPP is correlated with source noise

Together they are the minimal way of assessing phonation as non-modal (Garellek & Esposito, 2021; DiCanio, 2021; Garellek, 2022)

VOICE QUALITY- ACOUSTIC MEASUREMENTS

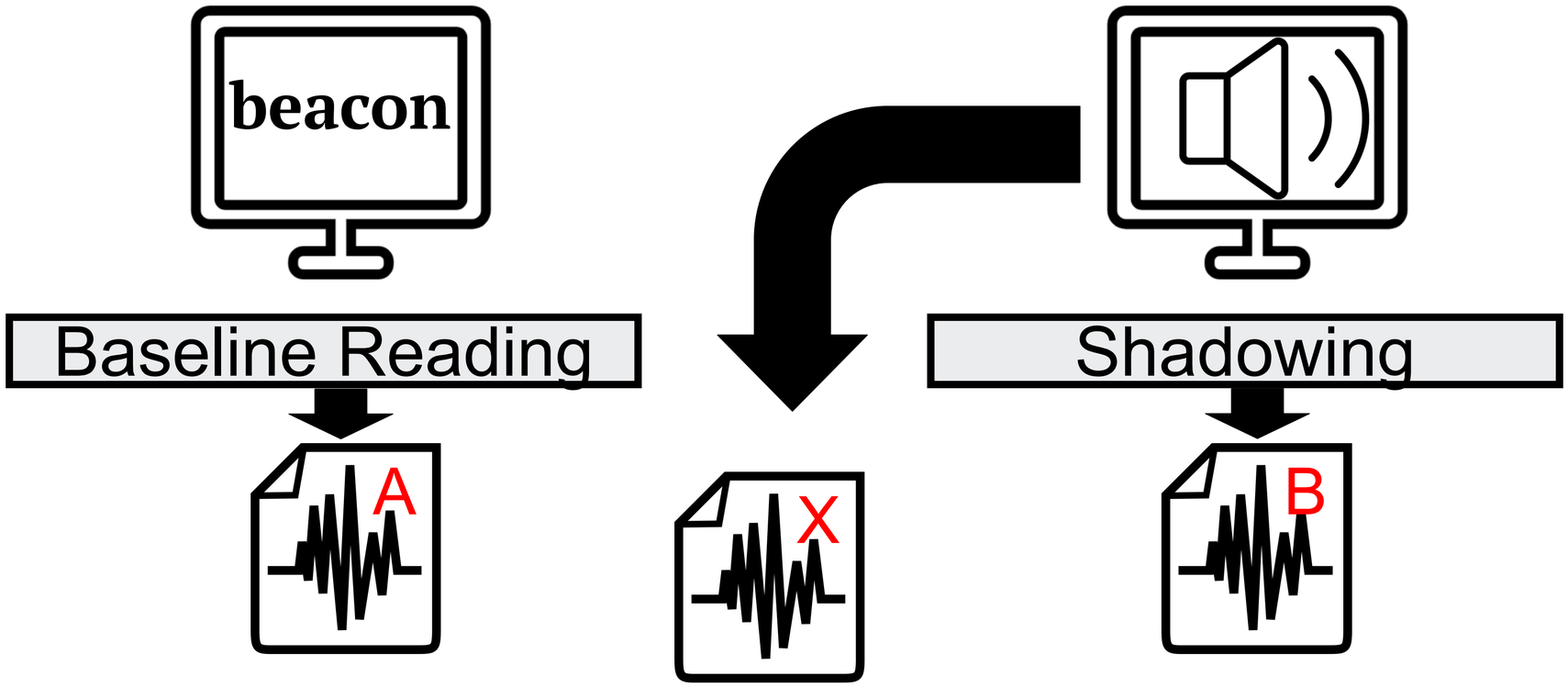
- Falsetto necessarily corresponds to a higher f0, while creaky and breathy (to a lesser extent) correlate with lower f0
- Generally, lower H1-H2 indicates creaky, higher indicates breathy
- CPP is higher in modal voicing with clear harmonic structure, and reduced in non-modal voice qualities

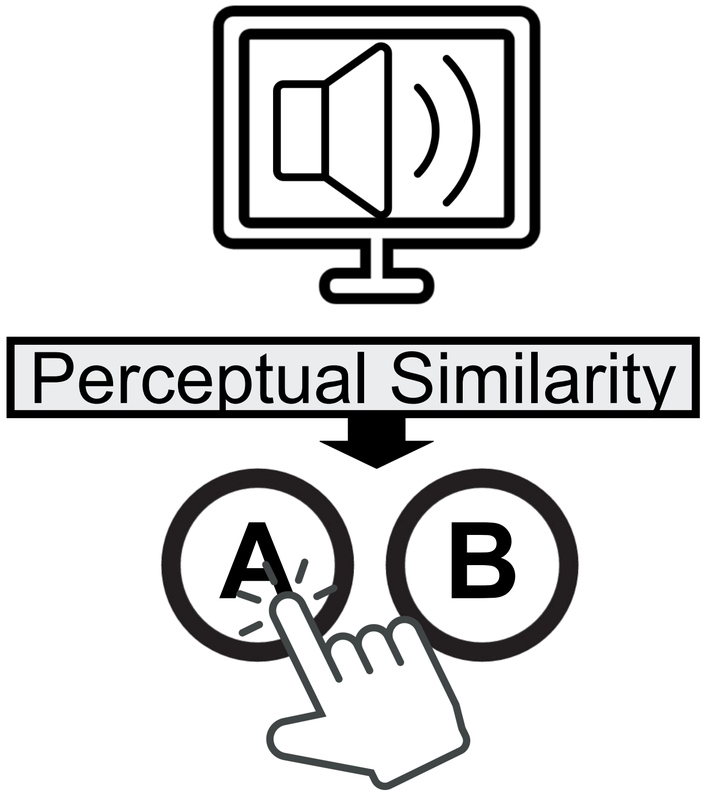
EXPLICIT IMITATION VS. IMPLICIT CONVERGENCE

When participants are explicitly asked to imitate what they hear, the degree of convergence increases vs.when they are asked to simply repeat the words (Clopper & Dossey, 2020, Schertz et al. 2023)

A potential difference being that the latter is assumed to be an automatic process, but the former seems to ask for reflection on salient features, and subsequent deliberative control over production.

DESIGN OF THE PRESENT STUDY




Perceptual Similarity
A
B

DESIGN OF THE PRESENT STUDY



PARTICIPANTS

These data come from 39 native English speaking undergraduate students (F=24,M=15) (age 18-24)

The data in this presentation are only from the 24 female participants

<u>Instruction Condition groups:</u>

- Implicit **('Repeat the word')**: n=20 (F=13, M=7)

- Explicit **('Repeat the word exactly as you heard the speaker say it')**: n=19 (F=11, M=8)

BASELINE READING

15 bisyllabic words:

beacon	novel
pigeon	bicep
staple	power
jelly	cushion
captain	vision
music	coffee
social	public
water	

These are shown on screen
in the baseline reading block

In the shadowing block,
presented as
audio recordings

SHADOWING

Audio Recordings of these 15 words were produced by 6 trained speakers (3 male; 3 female) across 4 voice qualities:

- -modal
- -creaky
- -breathy
- -falsetto

Each participant heard and repeated each of these 360 tokens exactly once in the shadowing block

ANALYSIS

To assess degree of convergence, two analyses are planned:

Perceptual judgments from an AXB Similarity Task

Acoustic measurements from the shadowing task

This presentation is focused only on the acoustic analysis

ANALYSIS

To assess degree of convergence, one option is to calculate difference in distance, relative to the stimulus item, between shadowed token and the baseline token as a dependent measure

(Babel, 2010, 2012; Clopper & Dossey 2020).

I consider f0 in terms of difference in distance in this analysis.

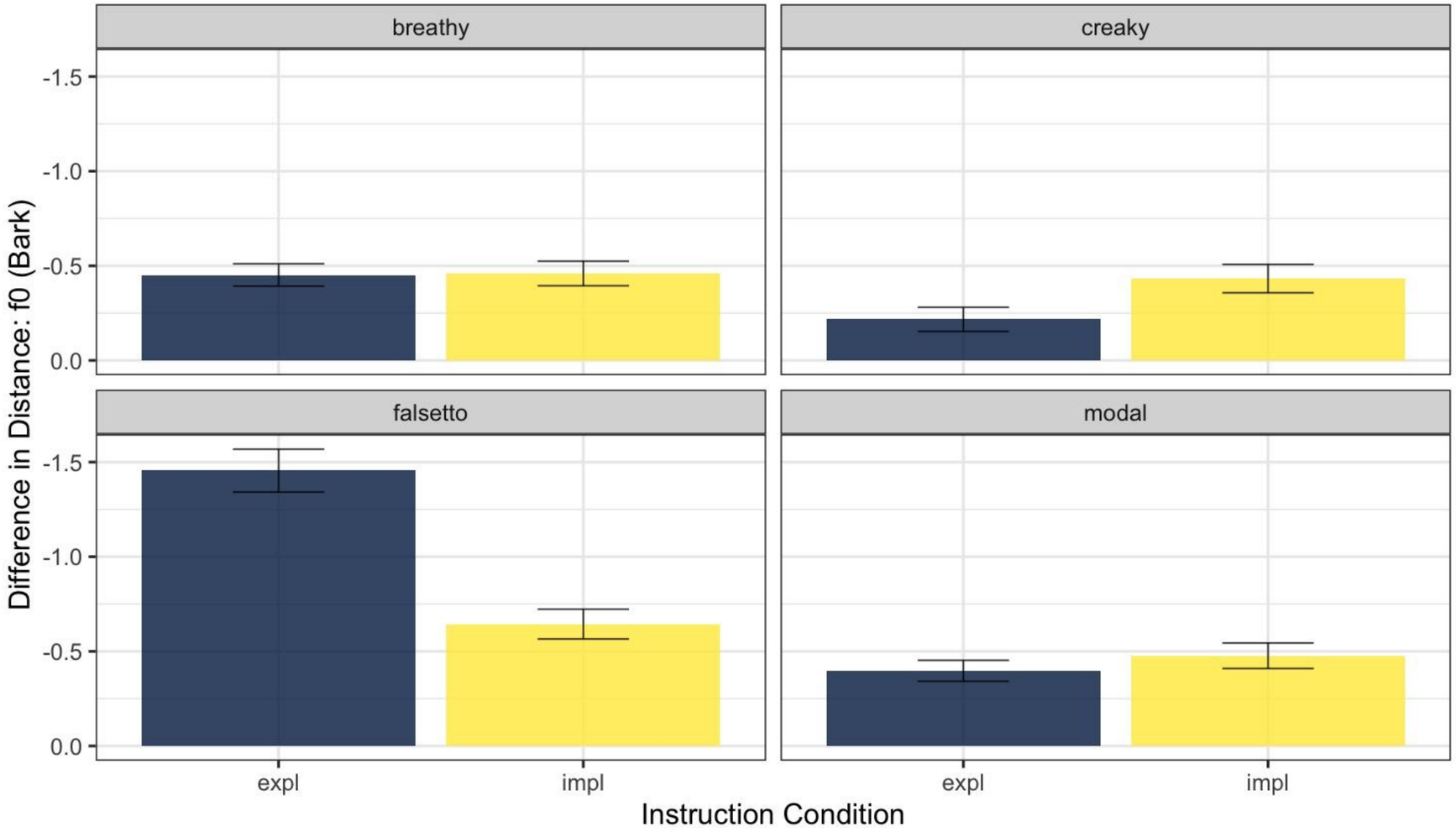
$$d(p,q) = \sqrt{\sum_{i=1}^{n} (q_i - p_i)^2}$$

ANALYSIS

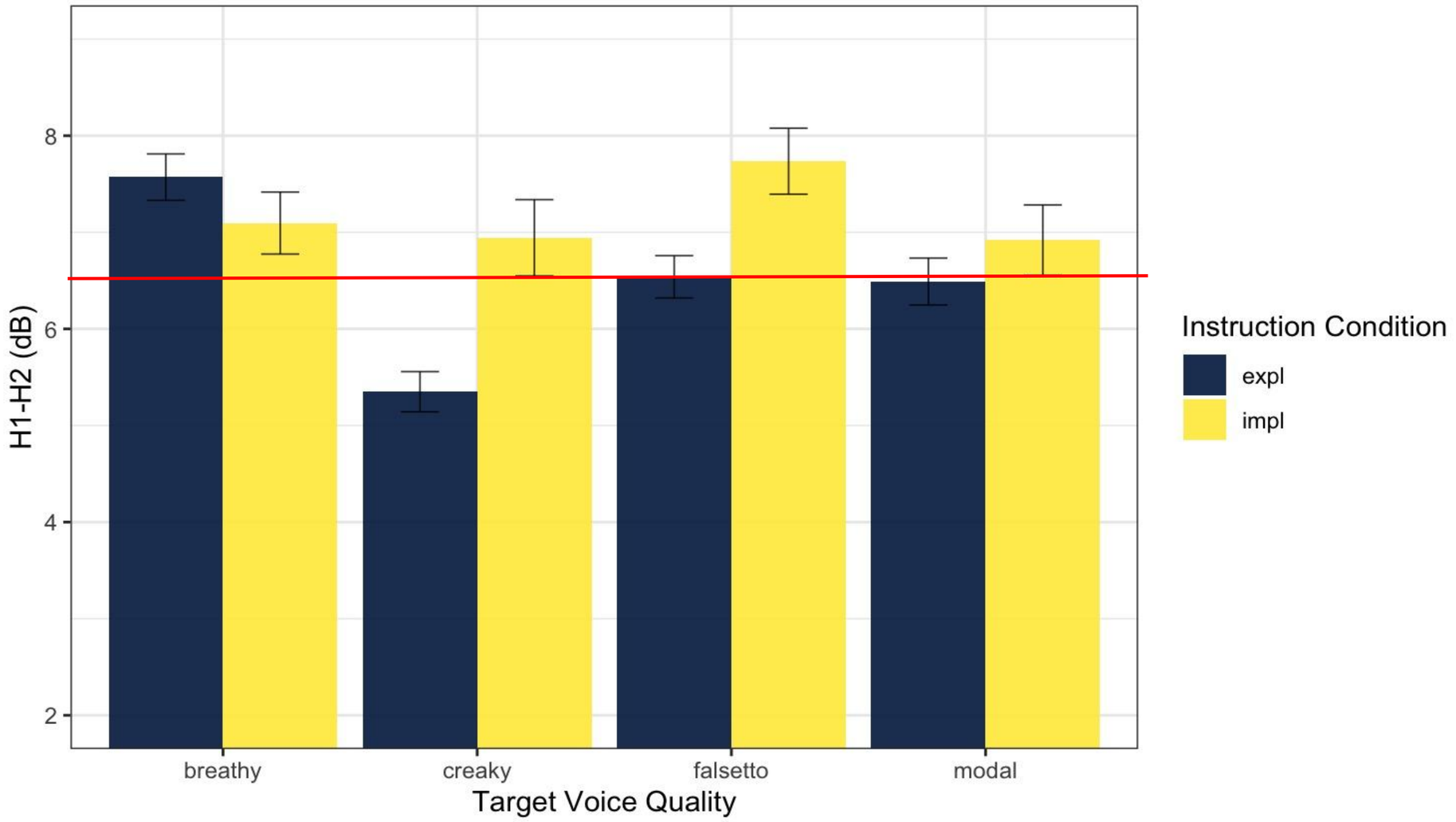
For H1-H2 and CPP, the measurements from the shadowed productions are used directly as the dependent measures.

- If a participant baseline is creakier (H1>H2) than a model talker, DID measures would require the participant become less creaky to show convergence to that talker

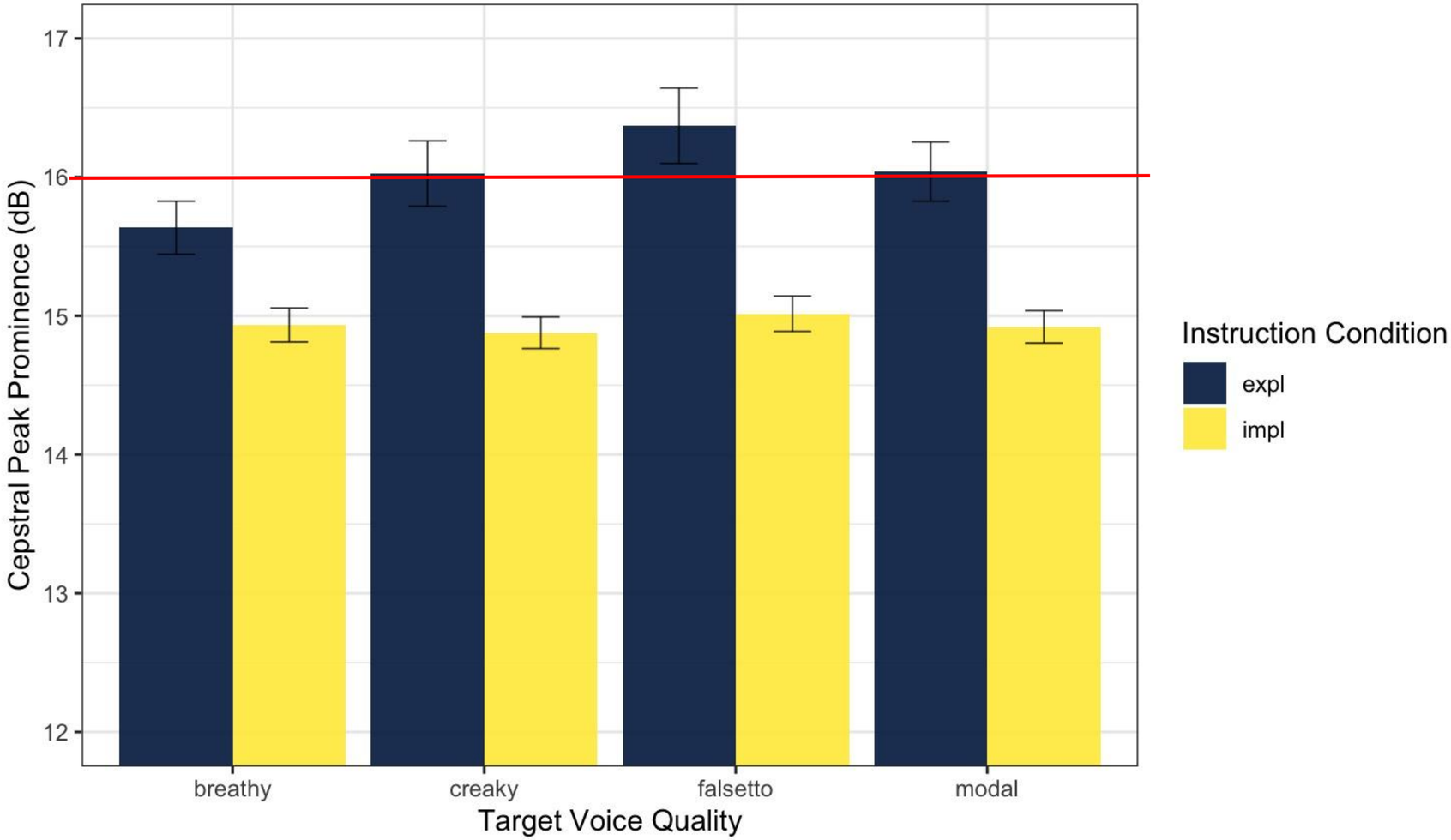
RESULTS-F0


breathy
creaky
falsetto
modal
-1.5
-1.0
-0.5
0.0
Difference in Distance: f0 (Bark)
expl
impl
Instruction Condition

RESULTS- H1-H2



RESULTS- CPP


17
16
15
14
13
12
Cepstral Peak Prominence (dB)
breathy
creaky
falsetto
modal
Target Voice Quality
Instruction Condition
expl
impl

RESEARCH QUESTIONS

Is voice quality a feature to which speakers will converge?

How will this differ under implicit vs. explicit instructions to imitate?

DISCUSSION- CONVERGENCE TO VQ

VQ seems to be imitated along these acoustic dimensions, generally

f0 convergence is greatest for falsetto, but changes in all voice qualities are significant

H1-H2 is distinct across creaky and breathy repetitions relative to baseline

CPP is on average lower for breathy and creaky repetitions relative to baseline indicating increased source noise expected of non-modal phonation

DISCUSSION-
EXPLICIT AND IMPLICIT INSTRUCTIONS

For f0, the expected pattern of greater convergence under explicit instructions is seen for falsetto, but the degree of convergence seems to decrease with explicit instructions elsewhere

H1-H2 change relative to baseline indicates convergence to creaky/breathy phonation only when instructions are explicit

CPP changes relative to baseline more under implicit instructions for all VQs (note: this is only true of female participants. Men show the opposite pattern)

DISCUSSION

People have varying degrees of control over phonation— various articulatory and acoustic parameters may or may not be equally controlled in production.

What is salient about the stimuli, and what is important to converge to seems to vary between instruction conditions. (e.g. female participants converge to f0 implicitly, always, but converge much more in terms of CPP explicitly)

CONCLUSIONS

Voice quality is converged to, on average, across these acoustic dimensions

In general, explicit instructions result in greater convergence, but they may also result in convergence to different acoustic features as compared to implicit instructions (cf .Schertz et al., 2023)

Complementary to studies describing quantitative differences in magnitude of convergence across implicit and explicit instruction conditions, these data also evidence qualitative differences in which acoustic dimensions are converged to most strongly (Dufour and Nguyen, 2013, Clopper & Dossey, 2020, Schertz et al., 2023)

FUTURE DIRECTIONS

- Ongoing AXB perceptual similarity task to evaluate convergence on a perceptual basis (Goldinger 1998)
  - Exploring other acoustic dimensions involved in the perception of VQ
  - Examining the ways in which gender of the listener and perceptions of the talkers' genders mediate these effects

SPEAKER RATING

To assess social perceptions of talkers, following the shadowing block participants complete an evaluation of each speaker:

Open ended:
-Gender
-Sexuality
-3 adjectives

Likert Scale (1-5):

- Speaker sounds feminine
- I found myself imitating this speaker

BONUS SLIDE ABOUT FAVE (ROSENFELDER, ET AL. 2014)

FAVE-align was used to auto-segment and align all audio recordings.
Sound files and text grids were then passed to FAVE-Extract

FAVE-Extract utilizes Praat's formant tracking algorithm and selects a set
of formant tracks estimated from Praat's Linear Predictive Coding (LPC) analysis.

These candidate measurements are compared to the distribution of measurements
for the same vowel quality in the
Atlas of North American English (ANAE)
and then re-evaluated based upon the distribution of the speakers' own productions of the
vowel in question

BONUS SLIDE
ABOUT REAPER

Robust Epoch And Pitch EstimatoR:(Liberman, 2015)
defines the local F0 estimates of a speech signal as inverse of time between consecutive glottal closure instant

shown to be accurate in its estimation of F0, and more effective at tracking F0 across very low pitch ranges than Praat with meaningful estimates being returned in ranges as low as 20 Hz

This was robust to ~97% of non-modal phonation tokens in these data

Reaper algorithm
- Signal Conditioning
- Feature Extraction
- Lattice Generation
- Dynamic Programming
- Backtrace and Output Generation

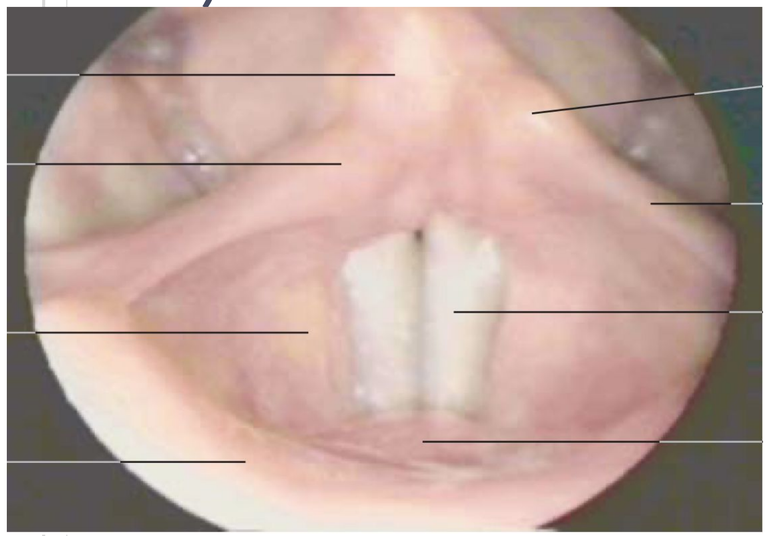
BONUS SLIDE ABOUT PRAATSAUCE

Praatsauce (Kirby, 2019) is a Praat implementation of and upgrade to VoiceSauc
(Shue et al., 2009) used to automate extraction of spectral measurements

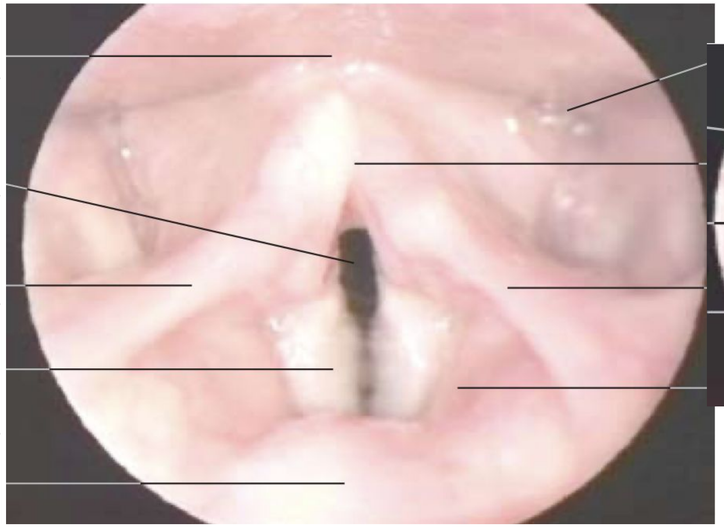
Of note:
H1-H2 are automatically corrected for formant influence
Spectral tilt is pitch-synchronous
A bevy of additional measurements are collected

It's not very good at pitch tracking in non-modal contexts, so I gave it f0 measurements and pitch tracks from reaper

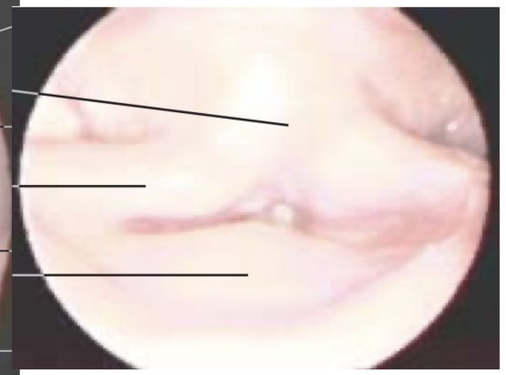
BONUS SLIDE VOICING & THE GLOTTIS (EDMONDSON & ESLING, 2006)

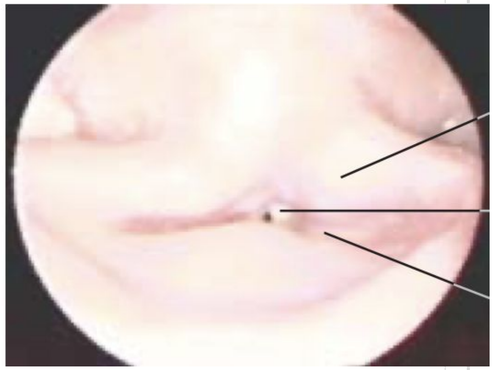


Mo
dal



Brea
thy





Cre
aky

BONUS SLIDE CPP

GARELLEK AND ESPOSITO (2021)

A cepstrum results from applying an inverse Fourier transform to an acoustic spectrum.
Allows examination of patterns, like the irregular glottal pulses of creak, in frequency spectra

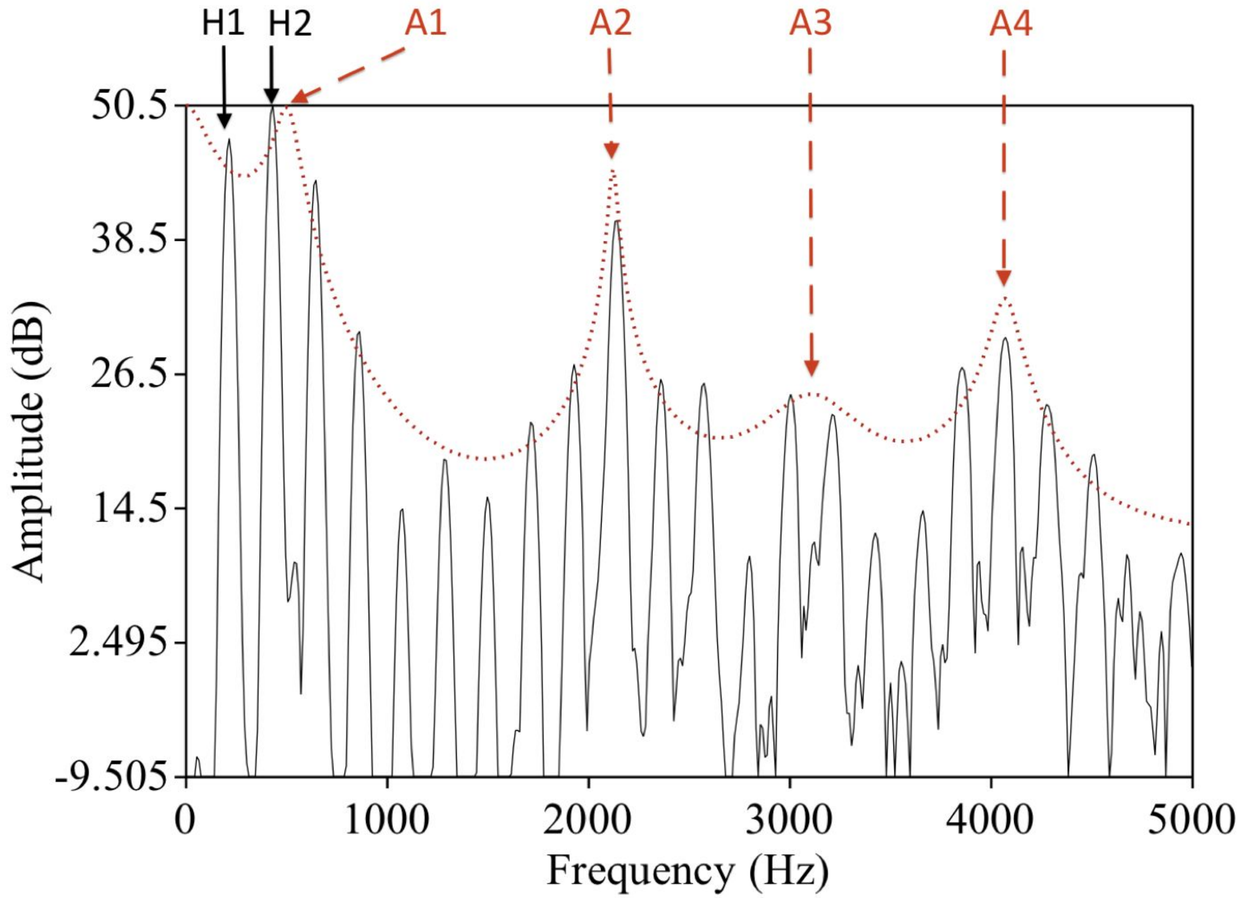
i. signal from time domain-> frequency
Ii. logarithm of the spectral amplitude
iii.transformation to frequency domain, where the final variable is on a time scale

Periodic harmonics in the spectrum are represented as a single large peak relative in the cepstrum.
CPP is the height of that peak relative to a regression line through the cepstrum (units are dB)

BONUS SLIDE H1-H2

DICANIO (2021)

Generally speaking, we expect lower spectral tilt values (positive slope) for tense/creaky phonation type and higher spectral tilt values (negative slope) for breathy phonation type



DISCUSSION- INTERACTIONS WITH GENDER

Female participants converge more strongly overall to VQ,

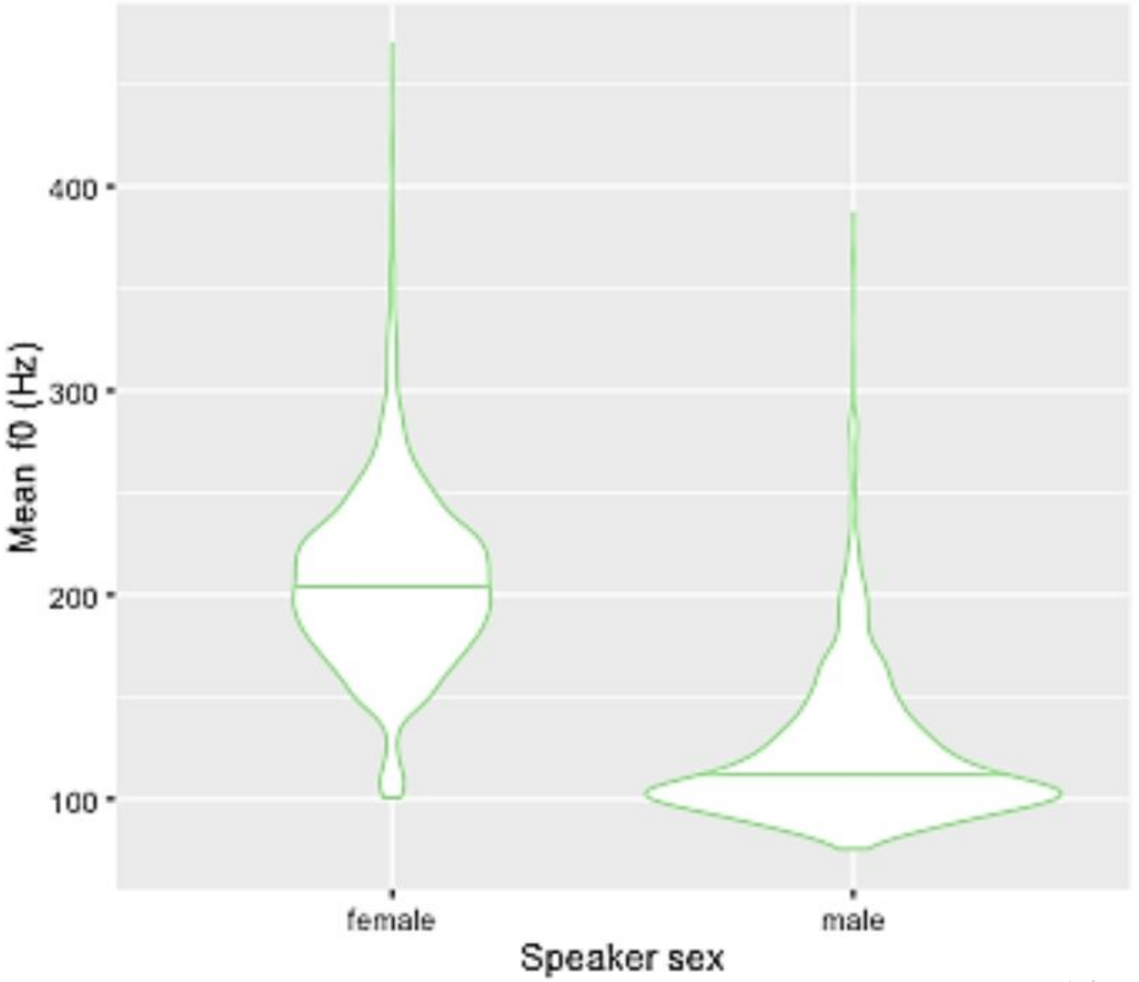
Male participants show more variation in convergence, accommodating most strongly to male talkers for H1-H2 and CPP, with much weaker patterns overall in implicit condition

In terms of f0 specifically, male participants converge very little on average to any talker under either condition, and their productions seem to be divergent in explicit instructions for creak and modal VQ

DISCUSSION- INTERACTIONS WITH GENDER

CPP change is greater in explicit condition for female participants, but smaller for male participants

This, and other gendered effects may be due to social selectivity related to the participant-talker gender pairing, but maybe also relative to what is salient in performance of different sexes/genders



Based on a figure based on a data set collected by Campbell-Kibler